



YAADCARD

PRINTING COST OPTIMIZATION

A3 + A4 + SEPARATE PACKAGING

A premium internal report for choosing the most efficient YaadCard deck size while preserving the duplex flashcard experience. Based on the supplied printer rates, current PDF layouts, and the proposed four-up packaging master.

Final recommendation

26-card lean deck: ₹67.75 | **36-card standard deck: ₹80.75**. The 36-card format gives the strongest balance of product size and cost per card.

Executive Summary

Two formats to test

Lean: 26 cards + packaging — 1 A3 duplex + 1 A4 duplex + 1/4 of a packaging master = ₹67.75.

Standard: 36 cards + packaging — 2 A3 duplex + 1/4 of a packaging master = ₹80.75.

The optimization is driven by a simple fact: every flashcard needs a front and a back, so the useful unit is a **finished duplex card**, not a single printed face. The separate packaging master then lets the tuck box be printed once per four decks instead of consuming a duplex back side.

Model	Cards	Card print	Packaging	Total	₹/card
Current baseline	30	₹76.00	embedded	₹76.00	₹2.53
Lean proposed	26	₹63.00	₹4.75	₹67.75	₹2.61
Standard proposed	36	₹76.00	₹4.75	₹80.75	₹2.24

Do not confuse total cost with cost efficiency. The 26-card deck is cheaper in absolute rupees, but the 36-card deck has a lower cost per card. The best business choice depends on whether the priority is a lower entry production cost or a stronger product/value ratio.

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Current YaadCard PDF Architecture

The supplied A3 editor describes an A3 landscape system with a standard 18-card duplex capacity. The first sheet is treated differently: it carries 12 cards plus the tuck-box net, while later sheets use 18-card pages. The print generator also adds one deck-index card automatically.

Current 30-card pack

The current implementation therefore represents 29 study cards + 1 index card. The first A3 sheet is the integrated “12 cards + packaging” sheet; its reverse contains the corresponding 12 backs while the inside of the box remains blank. A second A3 sheet carries the remaining 18 fronts and 18 backs.

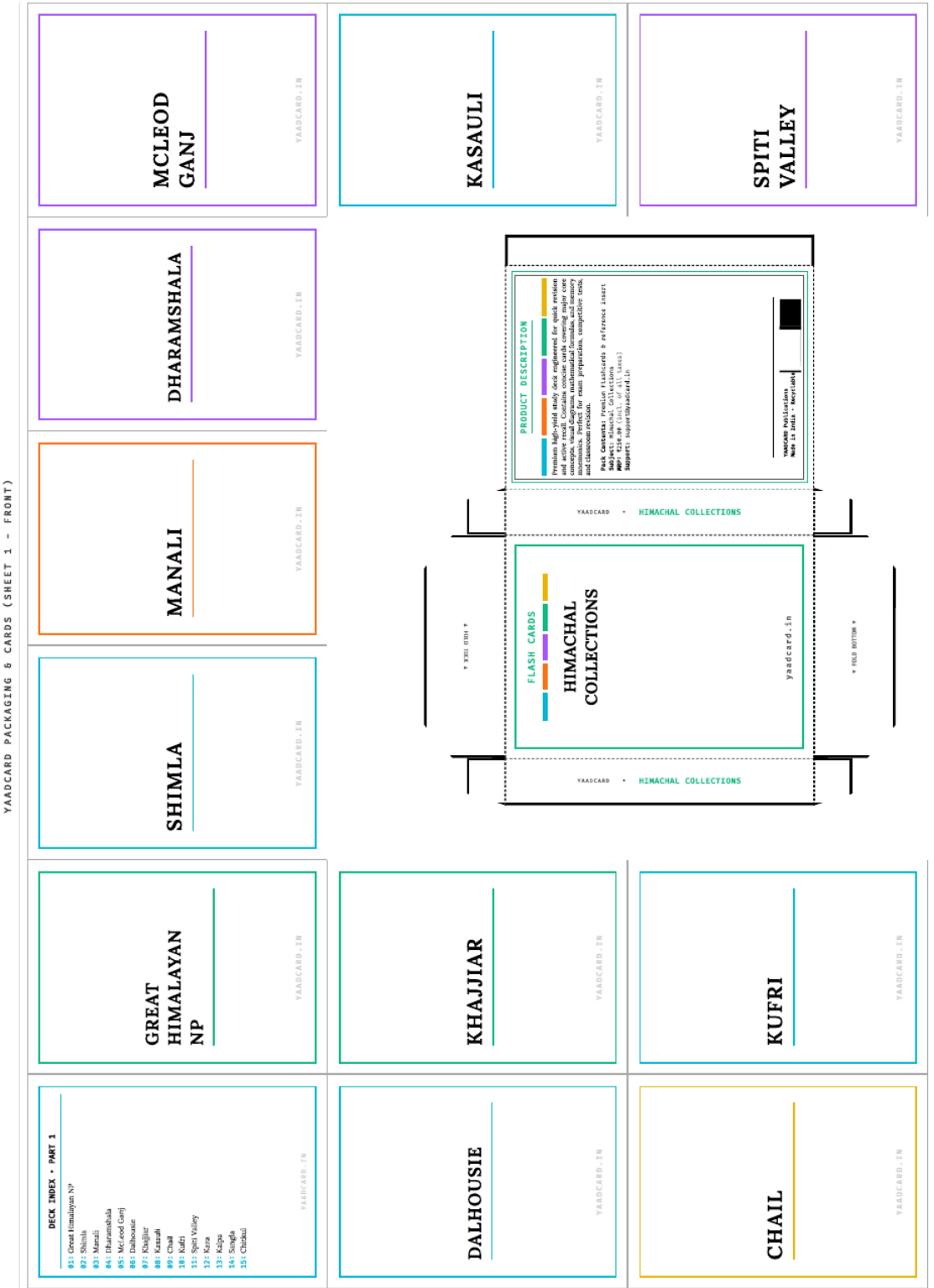
Supplied PDF, page 1: 12 card positions surrounding the tuck-box net.



Supplied PDF, page 1: same integrated packaging architecture.



Actual supplied layout — Himachal Collections



Supplied PDF, page 1: same first-sheet structure.

Printing Inputs

Printing component	Capacity	Cost	Effective rate
A3 single side — normal card side	18 card faces	₹19	₹1.06/face
A3 duplex — finished card sheet	18 finished cards	₹38	₹2.11/card
A4 duplex — finished card sheet	8 finished cards	₹25	₹3.13/card
A3 packaging master — single side	4 packages	₹19	₹4.75/package

Packaging unit cost = ₹19 / 4 = **₹4.75**

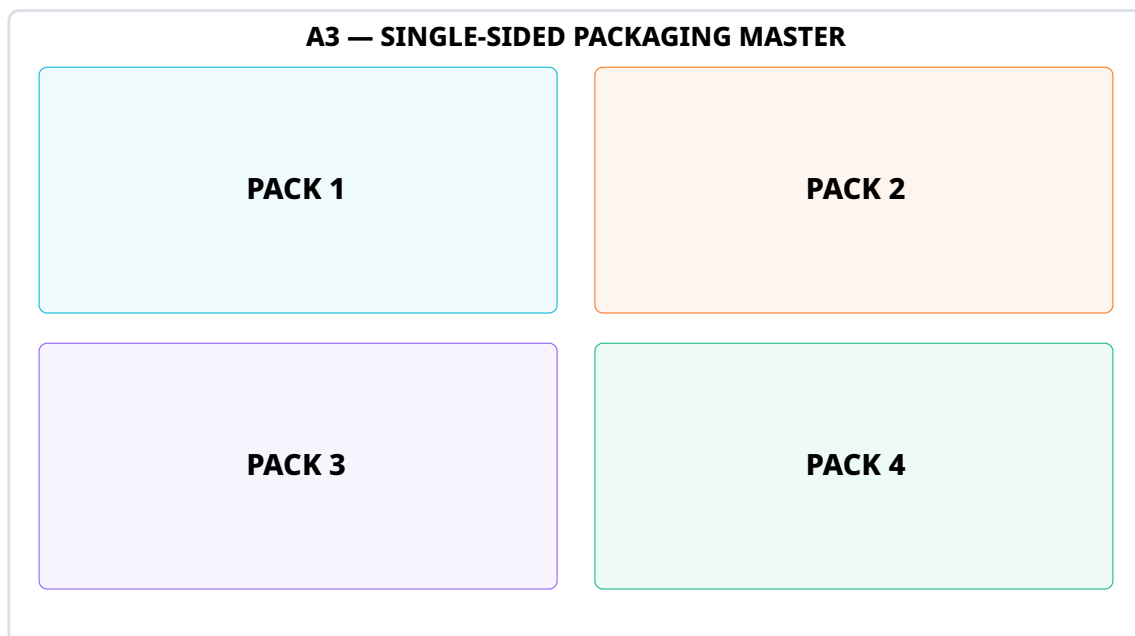
One A3 packaging master serves four finished decks and does not require a printed reverse side.

Why A3 should be the primary card format

A3 duplex produces 18 finished cards for ₹38, or about ₹2.11 per card. A4 duplex produces only 8 finished cards for ₹25, or ₹3.13 per card. Therefore A4 is useful mainly as a **bridge** when the target falls between A3 capacity blocks.

Proposed PDF Set

1. Packaging Master — NEW PDF



2. Standard A3 Duplex Card PDF



3. A4 Duplex Bridge PDF

The A4 editor is supplied as an A4 landscape version. For the cost model, the user’s measured capacity of 8 finished cards per duplex A4 sheet is used. It is a bridge format, not the preferred high-volume format.

Optimization Method

Let a be the number of A3 duplex card sheets and b the number of A4 duplex card sheets. For target n :

$$C(n) = 4.75 + \min\{38a + 25b : 18a + 8b \geq n, a, b \in \mathbb{Z}_{\geq 0}\}.$$

The model includes one-quarter of the packaging master per deck, assumes four packages share the ₹19 packaging sheet, and allows unused card capacity on a purchased sheet.

Full Optimization: 18–100 Cards

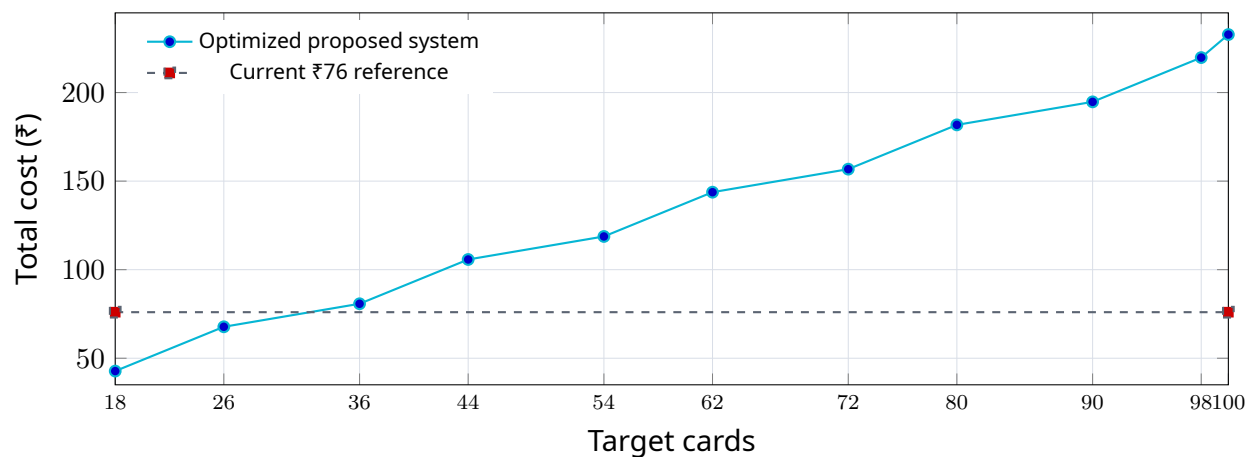
Every target from 18 through 100 is covered by the following contiguous ranges. Within each row, the same sheet combination is the cheapest solution for every target in that range.

Target range	A3 duplex	A4 duplex	Capacity	Total	₹/card at range top
18	1	0	18	₹42.75	₹2.38
19–26	1	1	26	₹67.75	₹2.61
27–36	2	0	36	₹80.75	₹2.24 at 36
37–44	2	1	44	₹105.75	₹2.40 at 44
45–54	3	0	54	₹118.75	₹2.20 at 54
55–62	3	1	62	₹143.75	₹2.32 at 62
63–72	4	0	72	₹156.75	₹2.18 at 72
73–80	4	1	80	₹181.75	₹2.27 at 80
81–90	5	0	90	₹194.75	₹2.16 at 90
91–98	5	1	98	₹219.75	₹2.24 at 98
99–100	6	0	108	₹232.75	₹2.33 at 100

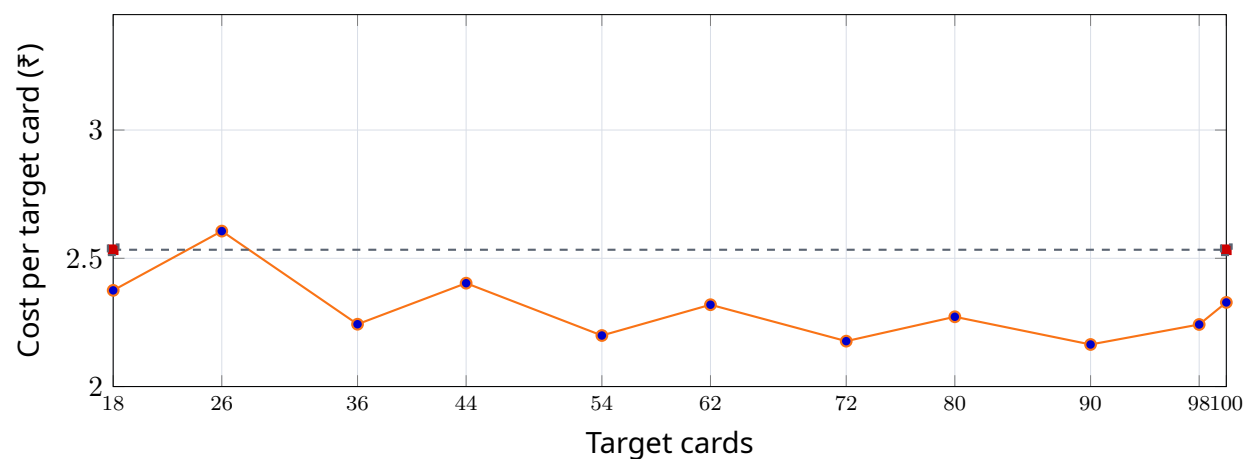
Important interpretation. The table is a “minimum spend for at least the requested number of cards” model. For example, targets 27 through 36 all require the same two A3 duplex sheets, so the total stays at ₹80.75 even though some capacity is unused.

Cost Curves

Total cost rises in sheet-sized steps



Cost per card generally improves as capacity is filled



The Two Best Practical Combinations

Option A — Lean 26-card deck

26 cards + separate packaging

1 A3 duplex = ₹38

1 A4 duplex = ₹25

1/4 packaging master = ₹4.75

Total = ₹67.75 ₹2.61/card

Compared with the current ₹76 reference, this saves **₹8.25 per deck** (10.9%). The compromise is that the cost per card is slightly higher than the current 30-card baseline because an A4 sheet is being used as a bridge.

Option B — Standard 36-card deck

36 cards + separate packaging

2 A3 duplex = ₹76

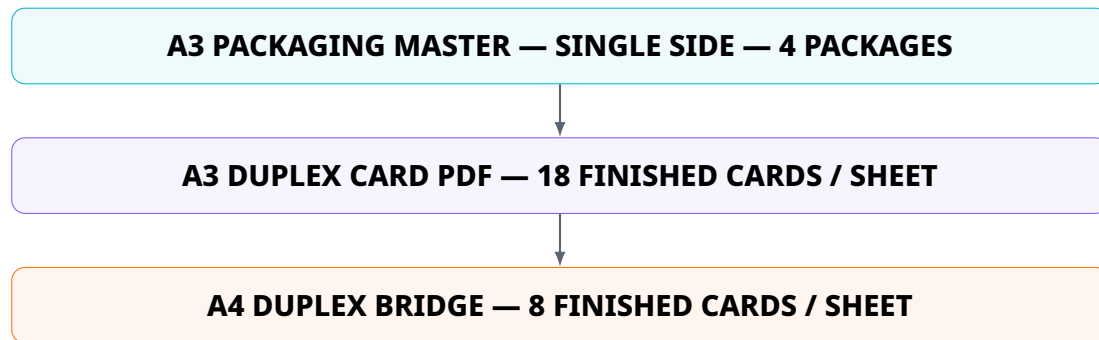
1/4 packaging master = ₹4.75

Total = ₹80.75 ₹2.24/card

Compared with the current 30-card / ₹76 reference, the new format gives **6 additional cards** for only **₹4.75 more total cost**. Cost per card falls by about **11.5%**.

	Current	26-card lean	36-card standard
Cards	30	26	36
Total production-print cost	₹76.00	₹67.75	₹80.75
₹/card	₹2.53	₹2.61	₹2.24
Change vs current	—	-₹8.25	+₹4.75
Card count change	—	-4	+6

Recommended Production Workflow



Files to maintain

1. **Packaging Master A3.pdf** — four packaging nets on one single-sided A3 page.
2. **Cards A3 Duplex.pdf** — 18 card fronts + 18 mirrored backs per duplex A3 sheet.
3. **Cards A4 Duplex.pdf** — 8 finished cards per duplex A4 sheet, used only when it improves the target fit.

Final Decision

Primary recommendation: 36-card standard deck.

It is the strongest practical compromise: it is materially larger than the current 30-card pack, uses only economical A3 duplex card sheets, and lowers the cost per card from about ₹2.53 to ₹2.24. The separate four-up packaging master adds only ₹4.75 per deck when batched in fours.

Secondary recommendation: 26-card lean deck.

Use this as a lower-cost format when production budget or retail price sensitivity matters more than maximum deck depth.

Before changing the production standard: physically test the new packaging master and duplex card PDFs with the printer. Verify front/back registration, cut lines, folding/gluing dimensions, paper thickness, and whether the printer's ₹19/₹38/₹25 rates remain unchanged for the new files.

Source Notes

- User-supplied rates used in this report: A3 single side ₹19; A3 duplex ₹38; A4 duplex ₹25.
- User-supplied capacities used: normal A3 18 cards; A4 duplex 8 cards; current integrated first-sheet area 12 cards + packaging.
- User-proposed packaging architecture: one A3 single-sided sheet carrying four packaging units, giving ₹4.75 packaging cost per deck.
- Supplied A3 JavaScript explicitly defines 18-card A3 duplex capacity and a first sheet with 12 cards plus the tuck box; it also automatically adds one index card to the printed deck.
- Supplied PDFs visibly demonstrate the current integrated packaging architecture.

- The business report records current production cost as ₹75, while the user's detailed printing calculation is ₹76. This report keeps the two figures separate and uses ₹76 for the printing comparison because that is the detailed rate provided for the current four-A3-side method.